

Automated quality assurance of segmentation techniques with statistically-based failure evaluation (SAFE)

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ABSTRACT

Whole brain segmentation and gray matter parcellation are of critical importance in neuroimaging pipelines. Ensuring the accuracy and quality of those results is important for drawing reliable conclusions, as well as evaluating processing techniques with downstream tasks. The contributions of this work are threefold: **(1)** to construct normative measures for brain region volumes that are distributable; **(2)** to propose the statistically-based failure evaluation (SAFE) method to accept or reject segmentations through statistical classifications; and **(3)** to compare the consistency of two brain segmentation/parcellation algorithms. We qualitatively validate SAFE on the SLANT and SynthSeg whole brain segmentation approaches by developing distribution statistics on 38,251 magnetic resonance images from 36 publicly available datasets. The experimental results indicate the efficacy of SAFE, with the method achieving 98% classification accuracy on 4,255 segmentations.

Keywords: neuroimaging, segmentation, statistical methodology, MRI

1. INTRODUCTION

Algorithms that segment brain structures into distinct tissue types, parcellate tissues into structural regions, and perform volumetric analyses can provide information about cognitive functions¹ and indicate the onset of neurodegenerative diseases.^{2–4} Quality assurance (QA) is a critical final step after processing when using segmentation results for scientific research or clinical management.^{5–8} Although manual QA strategies are the most reliable solutions for verifying segmentations,⁹ these approaches are time consuming and extremely difficult, even for experienced neuroanatomy experts. Methods like Mindcontrol¹⁰ aim to optimize this process but remain infeasible at scale.

Recent efforts dedicated to efficient QA have evaluated the robustness^{11,12} and reproducibility¹³ of segmentation algorithms. Although the work of Xiong *et al.*¹³ carefully evaluates segmentation variation across clinical acquisition protocols, the work does not inspect segmentation reproducibility on the same image. Many techniques present QA methods alongside segmentation algorithms. Bayesian QuickNAT¹⁴ estimates the segmentation uncertainty for quality control purposes, using four metrics to quantify the uncertainty. ReSeg¹⁵ is a deep neural network segmentation technique that accompanies reliability examinations and comparisons with other brain segmentation strategies. Our work addresses segmentation algorithms that lack an accompanying QA method and works to provide clarity about the effectiveness of two popular segmentation methods.

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^{1,2,3,4}See the acknowledgments.

1.1 Our Contributions

This work includes three main contributions. First, we construct a normative database (mean and standard deviation) of brain region volumes employing statistics generated on 38,251 publicly available magnetic resonance (MR) images. Our normative database of brain region volumes is distributable. We then propose the statistically-based failure evaluation (SAFE) approach, a binary classification method to accept or reject a brain segmentation based on the constructed normative values. This method is empirically validated on corrupted images using two algorithms: spatially localized atlas network tiles¹⁶ (SLANT) 3D whole brain segmentation and the SynthSeg¹⁷ segmentation tool, which is part of the FreeSurfer¹⁸ software package. Experiments involving images corrupted with artifacts indicate that SLANT is more robust to image distortions than SynthSeg. Finally, we conduct experiments to evaluate and compare the reproducibility of both algorithms when segmenting the same image multiple times. The results of this final contribution indicate that SynthSeg should be chosen to evaluate preprocessing tasks when reproducibility is crucial to ensure accurate results, unless additional effort is made to constrain random components of the SLANT method. The software associated with this project is available at <https://github.com/katembartz/SAFE>.

1.2 Statistical Theory

Throughout this analysis, we employ statistical methodology^{19,20} to compare and evaluate the SLANT and SynthSeg segmentation methods. For completeness, this section reviews standard statistical definitions and establishes notation.

First, we define two ways to classify numerical outliers. Tukey’s classification of box-plot statistics²¹ involves numerically sorting data points in ascending order. The first quartile (Q_1) is the value below which 25% of the data fall, and the third quartile (Q_3) is the value below which 75% of the data fall. That is, Q_1 and Q_3 are the values that mark the 25th and 75th percentiles of the data, respectively. The interquartile range (IQR) is their difference (ie. $IQR = Q_3 - Q_1$), which represents the spread of the middle 50% of the data. An outlier is a data point o such that $o < Q_1 - (1.5 \times IQR)$ or $o > Q_3 + (1.5 \times IQR)$.

Outliers can also be computed using second order statistics. Given a Gaussian distributed random variable with mean μ and variance σ^2 , 99.7% of the data is within the range $\mu \pm 3\sigma$. Anything beyond three standard deviations of the mean is a statistical rarity, which may warrant further attention. Though not universally defined, this classification can determine data outside of the normal distribution given mean and variance information.

Now, we will discuss how to estimate second order statistics with a collection of data observations. Let X_i and Y_i be random variables collected over the sample $i \in \{1, 2, \dots, n\}$. An unbiased estimate of the true population mean is the sample mean $\hat{\mu}_X$, and an unbiased estimate true population variance is the sample variance $\hat{\sigma}_X^2$, where both can be computed as

$$\hat{\mu}_X = \frac{1}{n} \sum_{i=1}^n X_i, \quad \text{and} \quad \hat{\sigma}_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu}_X)^2. \quad (1)$$

By the central limit theorem, given a large enough sample size n , the distribution of the sample mean is approximately normal. Thus, we may assume Gaussian statistics for a large sample size.

When comparing two random variables X and Y , the correlation coefficient ρ_{XY} is defined as

$$\rho_{XY} = \frac{\sigma_{XY}}{\sqrt{\sigma_X^2 \sigma_Y^2}}, \quad (2)$$

where σ_{XY} is the covariance, and σ_X^2, σ_Y^2 are the variances of X and Y , respectively. We can estimate ρ_{XY} by employing the definition for $\hat{\sigma}_X^2$ and $\hat{\sigma}_Y^2$ given in Eq. 1. Similarly, using an unbiased estimator of the covariance, we have

$$\hat{\sigma}_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu}_X)(Y_i - \hat{\mu}_Y). \quad (3)$$

This final relation estimates the covariance, and in combination with Eq. 1, an estimate of the correlation coefficient in Eq. 2 can be computed.

Finally, note the following important property: *Given independent random variables $X_1, \dots, X_m$, if $X = X_1 + \dots + X_m$, then $\mu_X = \mu_{X_1} + \dots + \mu_{X_m}$ and $\sigma_X^2 = \sigma_{X_1}^2 + \dots + \sigma_{X_m}^2$, where μ_X and σ_X^2 are the mean and variance of X , respectively.*

These statistical tools will be used to generative normative values and compare the SLANT and SynthSeg segmentation methods. To evaluate reproducibility, we calculate the coefficient of variation (CV) as

$$CV_X = \frac{\sigma_X}{\mu_X}, \quad (4)$$

where this value is estimated using the sample mean and sample variance estimates from Eq. 1.

2. MATERIALS AND METHODS

This paper generates normative measures for brain regions of interest (ROIs) segmented by SLANT (132 ROIs) and SynthSeg (98 ROIs). In an effort to increase robustness and reproducibility, the robust variant of SynthSeg version 2.0 and the SLANT-27 variant of SLANT version 1.2.0 were used for each respective segmentation.

2.1 Normative Measures

We collected 42,506 T_1 -weighted (T_1w) MR neuroimages from 38 publicly available databases. Table 1 lists the corresponding databases and number of unique subjects used in this work. Both sexes were represented, and the subjects ranged in age from 6 to 85+ years. The data included healthy controls and subjects with mild cognitive impairment, Alzheimer’s disease (AD), Parkinson’s disease, autism spectrum disorder, schizophrenia, and attention deficit hyperactivity disorder. Manual quality assurance was performed to exclude images with limited field-of-view (FOV), aggressive defacing, motion artifacts, noise, extreme atrophy, visual pathology, and other defects.²² The remaining images included 42,506 subjects. Although many databases include multiple scans per subject, only one image per subject was used to generate the normative measures to ensure the independence of the samples.

We divided the 42,506 images into a 90/10 training and testing split. We performed SLANT and SynthSeg segmentation on the 38,251 training images, and then calculated the volume of each brain region. As outlined in Figure 1(a), for each ROI, box-plot statistics were used to classify and remove outlier volumes, following the classification discussed in section 1.2. We calculated the mean and standard deviation of each ROI after the removal of the outlier values. This statistically analysis was done with R version 4.1.3.²³ The normative measures (computed mean and standard deviation) for all regions generated a model distribution database (MDD), where independent MDDs were created for both segmentation methods. The 4,255 testing images were reserved for evaluating the proposed SAFE method, which we outline in Section 2.2.

2.2 Statistically-based Failure Evaluation

SAFE is a binary method that takes two inputs: a MDD containing the mean volume and standard deviation for each ROI and a 3D segmented MR image volume. We statistically define in-distribution (ID) and out-of-distribution (OOD) following the Gaussian classification discussed in section 1.2, where ID includes data points within 3 standard deviations of the mean. As shown in Figure 1(b), this QA strategy is based on three verification steps. For each segmented ROI, SAFE first calculates the volume of the region. Then, SAFE normalizes the region volume relative to the total brain volume, which will henceforth be referred to as the ROI’s *brain occupancy*. As a first check, the method determines if these two measures for each ROI are ID or OOD. Some regions OOD are acceptable due to unaccounted for factors in the MDD, such as age, sex, and health status. Thus, SAFE proceeds to the next check if 95% of the ROIs are ID. Finally, SAFE checks for missing ROIs, where any zero volumes indicate segmentation failure and results in rejection of the segmentation. The SAFE method can be performed with any segmentation algorithm, and the results of this work include an analysis using the SLANT and SynthSeg methods.

Table 1: Database source and unique subjects (N) used in this study, as well as the breakdown for the train/test split by study. NADR²⁴ and SLEEP²⁵ study data is withheld from training to test out-of-distribution behavior.

Dataset Name	N	Train	Test
A4 ²⁶	1,742	1,584	158
ABIDE ²⁷	781	710	71
ABIDEII ²⁸	959	872	87
ADNI ²⁹	2,587	2,352	235
AIBL ³⁰	673	612	61
AOMIC-ID1000 ³¹	928	844	84
AOMIC-PIOP1 ³¹	216	196	20
AOMIC-PIOP2 ³¹	226	206	20
BHRC ³²	609	554	55
CCNP ³³	195	177	18
COBRE ³⁴	193	176	17
COGTRAIN ³⁵	166	151	15
CORR ³⁶	1,178	1,071	107
DLBS ³⁷	314	286	28
FCON1000 ³⁸	563	512	51
GSP ³⁹	1,570	1,427	143
HABS ⁴⁰	4,233	3,848	385
HBN ⁴¹	2,398	2,180	218
HCP1200 ⁴²	1,112	1,011	101
ICBM ⁴³	444	404	40
IXI	581	528	53
MCSA	1,799	1,636	163
MPILEMON ⁴⁴	226	206	20
NACC	3,680	3,346	334
NADR ²⁴	301	—	301
NARRATIVES ⁴⁵	315	286	29
NEUROPHENOM ⁴⁶	265	241	24
NIMHHRVD ⁴⁷	249	226	23
NKI ⁴⁸	1,314	1,195	119
OASIS3 ⁴⁹	1,340	1,218	122
OASIS4 ⁵⁰	655	596	59
PNC ⁵¹	1,600	1,455	145
PPMI ⁵²	1,975	1,796	179
QTIM ⁵³	1,199	1,090	109
SALD ⁵⁴	494	449	45
SCAN	4,702	4,275	427
SLEEP ²⁵	136	—	136
SLIM ⁵⁵	588	535	53
Total	42,506	38,251	4,255

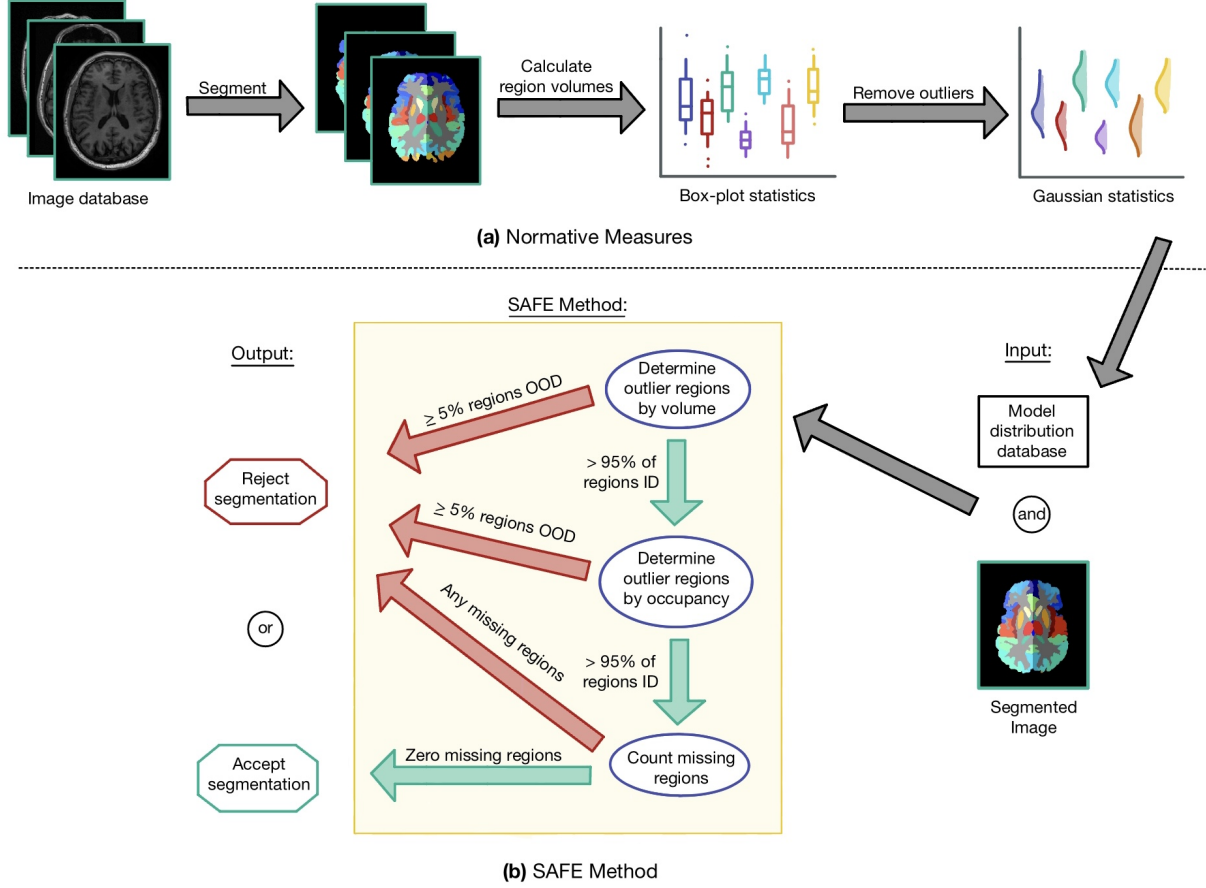


Figure 1: Method description. **(a)** Generation of normative measures for the model distribution database. Normative measures are computed for each region independently. **(b)** Proposed method: statistically-based failure evaluation (SAFE). SAFE takes as inputs a segmented MR image volume and the model distribution database, and outputs a decision to accept or reject the segmentation.

2.3 Reproducibility Evaluation

Brain segmentation is a common downstream task in medical image processing pipelines. Evaluating changes in volume regions can provide information about the effectiveness of certain preprocessing steps, such as harmonization^{56–58} and super-resolution.^{59,60} However, variations within segmented image output can influence these results, falsely indicating an error in a preprocessing step that is, in fact, due to some randomness component within the segmentation model. This is not a problem when using the SynthSeg method, as the algorithm is deterministic when the same computational environment and trained model is used. This can be controlled in scenarios where randomness would negatively influence results. However, the SLANT method produces nondeterministic inference due to a registration step within the model. The final contribution of this work is to determine the level of uncertainty this randomness generates.

3. EXPERIMENTS AND RESULTS

3.1 Normative Measures

All normative measures were generated using exactly one scan per subject to avoid bias, which allows us to treat each subject’s measures as an independent random variable. By the central limit theorem, each region volume is Gaussian distributed, where the mean and standard deviation characterize the data’s distribution.

Table 2 lists the normative volumes for the segmented subcortical regions generated from the SLANT and SynthSeg methods, as well as the segmented white matter and gray matter regions. SLANT uses the BrainCOLOR labeling protocol⁶¹ as a segmentation atlas. However, the SynthSeg method follows the FreeSurfer¹⁸ classification, based on a manually labeled training set.⁶² We follow the Mindboggle⁶³ conversion between the BrainCOLOR subcortex labels and FreeSurfer labels to properly discuss similar regions across the two methods. Table 2 compares regions represented in both the SynthSeg and SLANT method outputs.

To compare the SLANT and SynthSeg normative volumes, we computed the correlation coefficient ρ intra-region, using the unbiased estimators described in Section 1.2. The computed coefficient indicates positive correlation across all regions, though not near the value 1 that indicates a perfect positive linear relationship. This reveals differences between the segmentation methods, likely due to the algorithm’s training data and specific region definitions inherent to the models. Volumetrics generated from segmented images inform clinical diagnostics. Thus, statistical differences in region volumes across segmentation algorithms can lead to naive conclusions if not accounted for and understood properly. Because of these region differences, we built a MDD for both methods, which are provided in the publicly available code associated with this project at <https://github.com/katembartz/SAFE>.

Even though cortical parcellation was performed with both the SLANT and SynthSeg methods, normative measures for the cerebral gray matter were calculated by adding the independent regions. Thus, the mean and variance for the entire gray matter are the sum of each region’s mean and variance, respectively. Due to the number of estimations leading to approximations of the true data, we do not compute the correlation coefficient for the full cerebral gray matter tissue regions.

3.2 SAFE for automated QA

Due to ventricle enlargement that accompanies age and other diseases,⁶⁴ the six regions that encompass the ventricles—3rd ventricle, 4th ventricle, right and left lateral ventricle, and right and left inferior lateral ventricle—were excluded from the SAFE method. The remaining 126 brain regions for SLANT and 92 brain regions for SynthSeg informed the statistical measures stored in the MDD, following the procedure described in section 2.1.

QA is important to classify and remove algorithmic failures. Using the 4,255 testing images from the 38 databases (as listed in Table 1), we simulated 8 types of corruption to encourage cases in which the segmentation methods fail. The corruptions, corresponding parameters, and number of images are shown in Table 3, where the artifact simulations followed the structure of Hays *et al.*⁶⁵ All 38 databases were represented in each of the image corruption types. Artifacts ranged from mild to severe and were randomly simulated using the TorchIO⁶⁶ library from Python. Anisotropic thick-slice images were generated using the radifox-utils package,⁶⁷ which differ from the artifact random anisotropy by the way in which downsampling is performed. The random anisotropy artifact from the TorchIO library performs a downsampling step using nearest neighbor interpolation, followed by linear interpolation to upsample to the original resolution. The anisotropic thick-slice corruption employs a more convoluted downsampling step that closer reflects how MR images are acquired. It uses a Shinnar-Le Roux algorithm to generate a rect-like slice profile that first performs a discrete convolution within the image volume. The volume is then subsampled with cubic B-spline FOV-aware resampling. The resulting image is not upsampled to the original resolution. This subtle difference between the two anisotropy corruptions produces different segmentation results.

All corrupted images were independently segmented with SLANT and SynthSeg, followed by automated QA with SAFE. SLANT includes N4 bias field correction, intensity normalization, and registration as preprocessing steps. Thus, SLANT is robust to minor artifacts, but segmentation fails for more extreme corruption. The SynthSeg algorithm includes a preprocessing step to resample input images to 1mm resolution using trilinear interpolation. Thus, SynthSeg is highly robust to the anisotropic thick-slice images that reflect low resolution acquired MR images. However, SynthSeg performed poorly on images containing artifacts, and in many cases, failed to produce any type of segmentation. Figure 2 shows corrupted images in which SAFE rejected the segmentation. These results reveal how even in cases where the input image is severely corrupted, the SLANT segmentation still resembles a brain, while SynthSeg fails to produce a brain-like image segmentation on the same corruption.

Table 2: Normative mean volume measures (standard deviation) in mm^3 for segmented tissue regions generated from SLANT and SynthSeg segmentation methods across various brain ROIs. The correlation coefficient (ρ) was also computed to compare the segmentation methods across ROIs.

ROI	SLANT (mm^3)	SynthSeg (mm^3)	ρ
3rd Ventricle	2,234.18 (± 818.00)	1,046.43 (± 518.29)	0.2388
4th Ventricle	1,779.74 (± 611.23)	1,557.67 (± 432.95)	0.0158
Right Accumbens Area	419.16 (± 107.46)	723.32 (± 138.84)	0.3932
Left Accumbens Area	509.15 (± 118.16)	692.39 (± 119.53)	0.3486
Right Amygdala	1,052.74 (± 143.52)	1,884.81 (± 238.66)	0.0940
Left Amygdala	1,101.02 (± 156.44)	1,751.78 (± 226.61)	0.0637
Brain-stem	18,544.59 ($\pm 2,224.53$)	23,242.83 ($\pm 2,359.20$)	0.0427
Right Caudate	3,331.63 (± 571.76)	3,972.46 (± 559.52)	0.3304
Left Caudate	3,330.38 (± 559.92)	3,752.67 (± 556.92)	0.3571
Right Cerebellum Cortex	54,109.50 ($\pm 6,691.10$)	56,386.85 ($\pm 6,178.28$)	0.1142
Left Cerebellum Cortex	53,394.91 ($\pm 6,684.64$)	58,210.46 ($\pm 6,337.39$)	0.1293
Right Cerebellum White Matter	11,570.66 ($\pm 1,750.40$)	13,156.59 ($\pm 1,666.89$)	0.0847
Left Cerebellum White Matter	12,131.37 ($\pm 1,830.69$)	13,853.66 ($\pm 1,767.08$)	0.0926
Right Cerebral White Matter	192,752.76 ($\pm 26,669.40$)	221,926.88 ($\pm 26,022.59$)	0.1766
Left Cerebral White Matter	189,624.11 ($\pm 26,850.94$)	225,614.23 ($\pm 26,543.59$)	0.1723
Right Hippocampus	4,025.36 (4 ± 87.87)	3,857.25 (± 429.80)	0.0738
Left Hippocampus	3,815.35 (± 462.58)	3,979.30 (± 452.69)	0.0944
Right Inferior Lateral Ventricle	374.03 (± 228.50)	429.14 (± 144.48)	0.0645
Left Inferior Lateral Ventricle	357.61 (± 210.88)	400.65 (± 126.36)	0.0973
Right Lateral Ventricle	12,196.85 ($\pm 7,546.31$)	10,377.23 ($\pm 6,614.35$)	0.3399
Left Lateral Ventricle	12,631.73 ($\pm 7,587.44$)	10,907.97 ($\pm 7,385.83$)	0.3481
Right Pallidum	1,337.86 (± 193.07)	1,677.01 (± 233.87)	0.2205
Left Pallidum	1,518.57 (± 199.91)	1,484.66 (± 251.29)	0.2391
Right Putamen	4,303.90 (± 644.03)	5,312.41 (± 752.74)	0.3430
Left Putamen	4,447.34 (± 669.58)	5,482.66 (± 739.39)	0.3529
Right Thalamus	7,823.20 (± 941.66)	7,275.49 ($\pm 1,015.68$)	0.2711
Left Thalamus	7,795.84 (± 957.19)	6,994.54 (± 961.77)	0.2796
Right Ventricle DC	4,711.44 (± 532.63)	4,164.78 (± 487.86)	0.1524
Left Ventricle DC	5,092.84 (± 584.05)	4,178.63 (± 490.59)	0.1380
Right Cerebral Gray Matter*	322,206.02 ($\pm 58,051.03$)	268,847.98 ($\pm 40,003.64$)	—
Left Cerebral Gray Matter*	323,892.38 ($\pm 59,175.77$)	270,084.39 ($\pm 40,685.55$)	—

*Gray matter volumes computed as sums and no correlation coefficient computed.

The confusion matrices in Table 4 and Table 5 display the performance of SAFE on the 4,255 images segmented with SLANT and SynthSeg, respectively, compared to the ground truth manual evaluation. Manual QA directly compared the input image with the segmented image and focused on brain alignment and anatomical accuracy. These results reveal that SAFE performed with 98.1% accuracy on images segmented with the SLANT method, and 98.9% accuracy on images segmented with the SynthSeg tool. SAFE is an effective automated method to determine segmentation failures, which can indicate catastrophic input images corrupted by artifacts or other preprocessing errors.

3.3 Reproducibility Experiments

To provide clarity on the inherent randomness present in the SLANT method, we conducted an experiment to evaluate the reproducibility of the algorithm intra-image and intra-region. Figure 3 outlines this procedure: independent segmentations are repeated for the same image, yielding potential changes between region volumes

Table 3: Types of image corruptions used to validate the SAFE method. The testing set from Table 1 was divided across each corruption and the number of subjects (N) are listed accordingly.

Image Corruption	N	Parameters (Ranges)
None	2,199	Original Image
Artifact: Random Anisotropy	302	<code>downsampling</code> $\sim \mathcal{U}(15, 25)$
Artifact: Random Bias Field	301	<code>coefficients</code> $\sim \mathcal{U}(2, 5)$
Artifact: Random Blurring	297	$\mathcal{N}(0, \mathcal{U}(25, 100))$
Artifact: Random Ghosting	294	<code>num_ghosts</code> $\sim \mathcal{U}\{2, \dots, 6\}$, <code>intensity</code> $\sim \mathcal{U}(1, 10)$
Anisotropy Thick-Slice	289	7mm slice thickness with 1.75mm slice gap
Artifact: Random Motion	287	<code>translation</code> $\sim \mathcal{U}(15, 20)$, <code>num_transforms</code> = 30, <code>degrees</code> $\sim \mathcal{U}(10, 170)$
Artifact: Random Noise	286	0-mean, standard deviation $\sim \mathcal{U}(8k, 10k)$

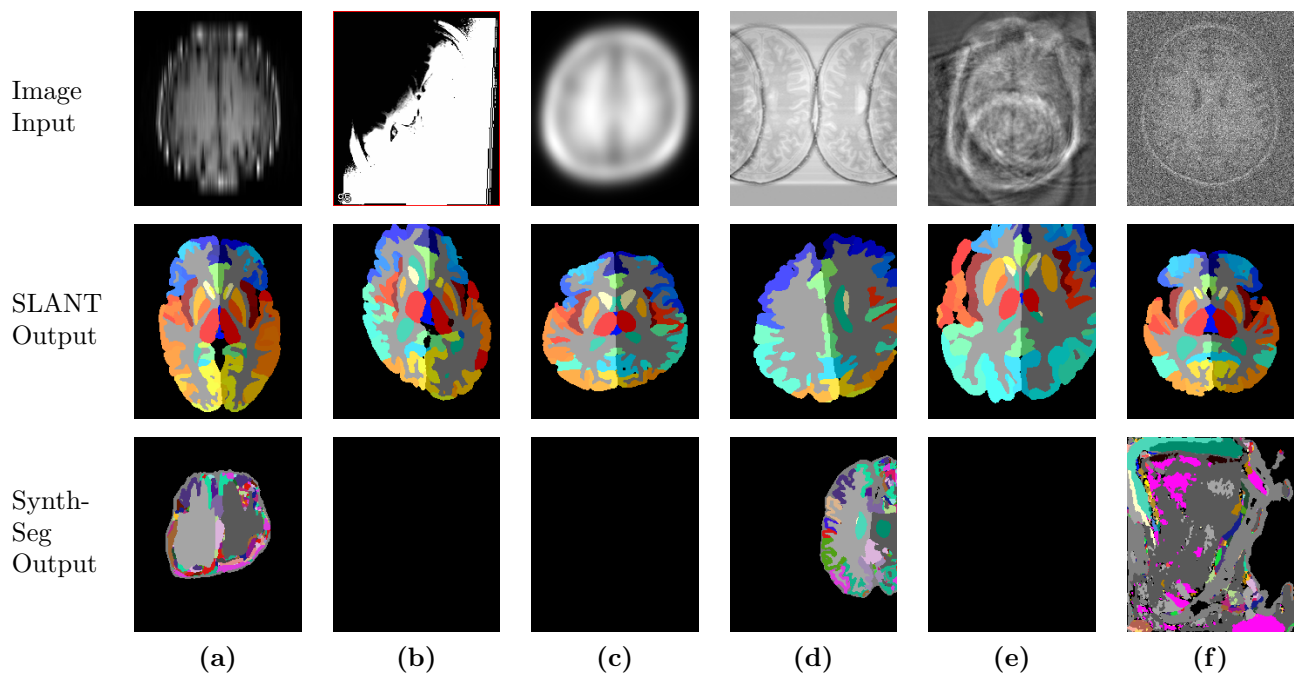


Figure 2: SLANT and SynthSeg segmentations rejected by the SAFE method. Artifacts: (a) random anisotropy; (b) random bias field; (c) random blurring; (d) random ghosting; (e) random motion; (f) random noise. Blank images indicate failure of the algorithm to segment any regions of the brain.

Table 4: A confusion matrix representation for the experimental results of SAFE for SLANT segmentations with sensitivity (Sens), specificity (Spec), precision (Prec), negative predictive value (NPV), and accuracy (Acc).

		SAFE		
		Accepted	Rejected	
Manually	Accepted	2247	19	Sens: 99.16 %
	Rejected	63	1926	Spec: 96.83 %
		Prec: 97.27 %	NPV: 99.02 %	Acc: 98.07 %

from one segmentation to the next. This can be quantified by calculating the intra-region CV across the various segmentations for the given input image.

Table 5: A confusion matrix representation for the experimental results of SAFE for SynthSeg segmentations with sensitivity (Sens), specificity (Spec), precision (Prec), negative predictive value (NPV), and accuracy (Acc).

		SAFE		
		Accepted	Rejected	
Manually	Accepted	2380	41	Sens: 98.31 %
	Rejected	5	1824	Spec: 99.73 %
		Prec: 99.79 %	NPV: 97.80 %	Acc: 98.92 %

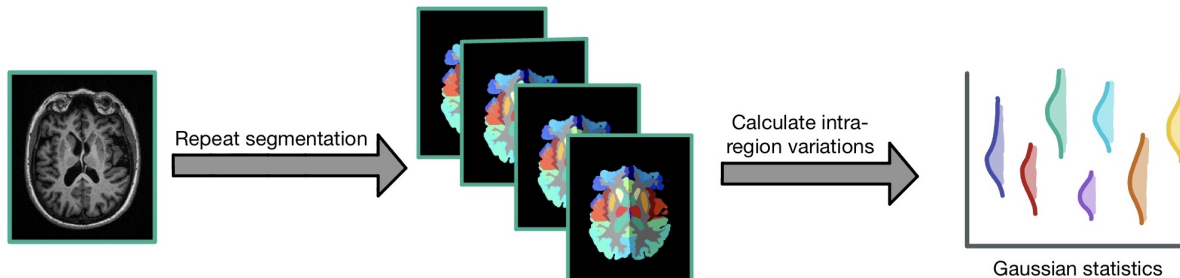


Figure 3: Experimental procedure for reproducibility experiment.

Five test images from each of the 38 databases in Table 1 were selected for the reproducibility experiment, totaling 190 images. These images were segmented with SLANT 30 times, where each segmentation was independent from the previous. We then calculated the intra-region variations by computing the CV between each region for 30 segmentations performed on the same input image. We performed this same experiment using SynthSeg segmentation, and our results confirmed the deterministic nature of the method; there was no variation between region volumes over multiple segmentations.

Figure 4 displays the calculated CV for select SLANT regions across the 190 images. For regions with a left-right hemisphere pair, only the right hemisphere regions are depicted in the graph. These results indicate nonzero randomness within the SLANT segmentation method due to the random seed within the registration step. In outlier cases, the CV exceeded 0.10, where SLANT performed more consistently for larger regions. This may in part be due to the discretization of segmented images with 1mm resolution; in regions with lower volume, varying by a few pixels causes more variation across the total volume compared to larger regions.

Notably, the regions with the highest CV were the left and right inferior lateral ventricle. This same result was observed by Xiong *et al.*¹³ where their work tested SLANT segmentation variations across various clinical protocols rather than within the same image. The results of this experiment suggest that SLANT may not be a preferred segmentation method in cases where region volume comparisons are used to evaluate preprocessing steps without additional effort to lock random components. In these scenarios, the deterministic nature of the SynthSeg method will provide more clarity of comparison.

4. DISCUSSION

When comparing like regions between the SLANT and SynthSeg segmentation methods, the normative measures generated are only slightly correlated. In some regions, the difference between the mean values is striking, as displayed in Table 2. Specifically, the computed mean volume for the 3rd ventricle region is over twice as large when segmented using SLANT than when using SynthSeg.

Despite efforts to ensure independence in samples, there are a few factors to consider that influence this analysis. Firstly, we assume that segmented regions are independent from one another. However, if SynthSeg classifies region 1 as having less volume than the SLANT classification, there will exist some region 2 in which SLANT assigns a larger volume than SynthSeg. Small differences in how the two models are trained and define region boundaries can lead to significant differences between the region volumetrics.

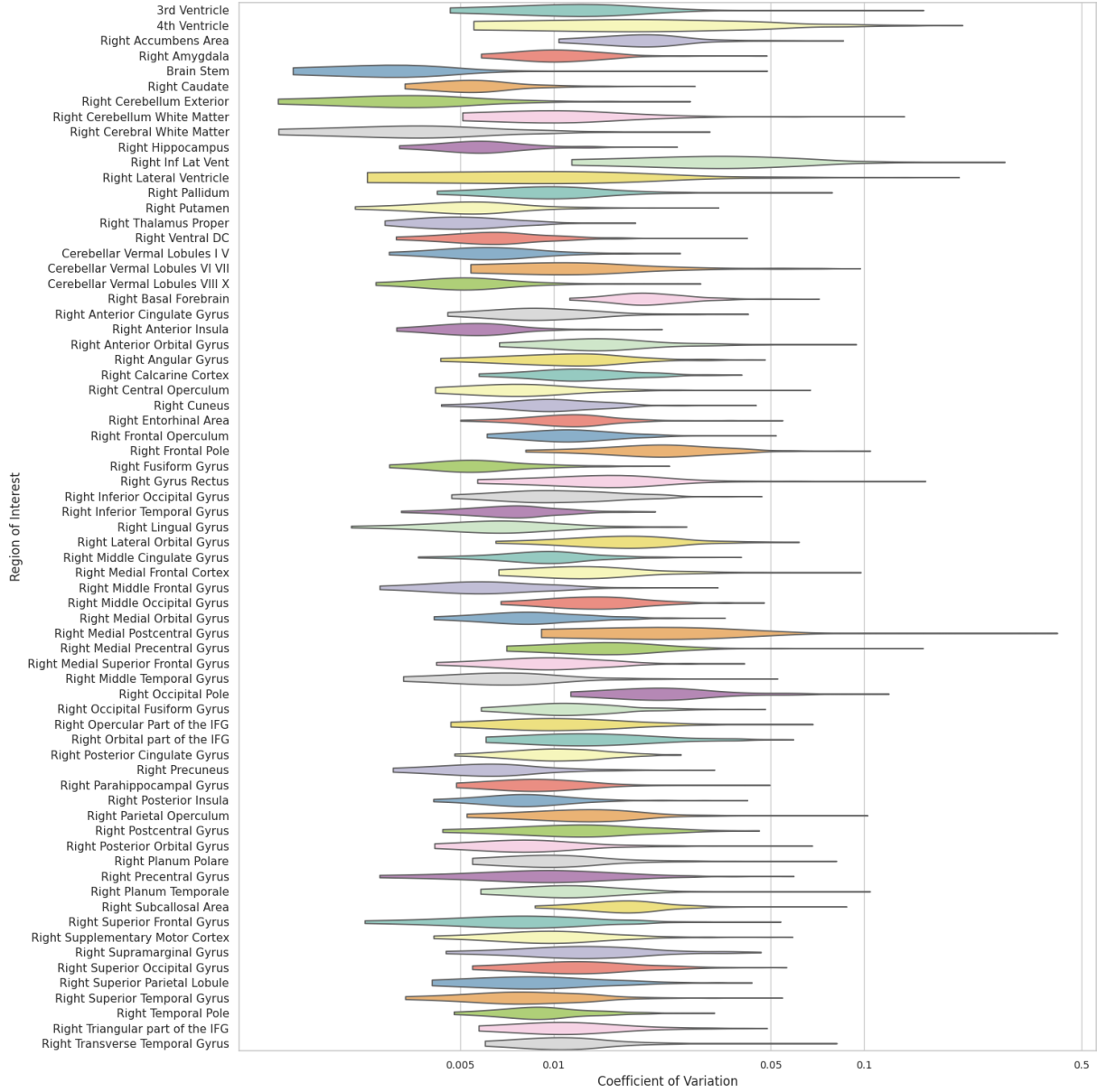


Figure 4: Violin plot depiction of coefficient of variation across 190 images for SLANT reproducibility evaluation. For regions with a right-left hemisphere pair (e.g. right/left cuneus), only the right hemisphere regions are displayed. Note the use of a logarithmic scale to better display the CV data, as outliers are skewed.

Secondly, upon visual inspection, the probability density functions for select regions do not follow a Gaussian distribution curve. This is especially apparent when analyzing the ventricle regions. As discussed, ventricle enlargement is a natural process that accompanies age and other diseases. Because this analysis does not account for demographic data, an effect of this sort can skew how the correlation is computed as well as the effectiveness of using second order statistics to classify outliers.

5. CONCLUSION

QA of whole brain segmentation is crucial to verify medical image processing pipelines, and manual techniques are often impractical. This work proposed SAFE, an automated quality assurance method to accept or reject

whole brain segmentations. Building statistics on a large database of 38,251 images, the method was shown to be robust to image artifacts and corruption. SAFE performed with over 98% accuracy on a test set of 4,255 brain images with a wide variety of simulated corruptions segmented with SLANT and SynthSeg.

Due to the general and statistical nature of our method, SAFE is most successful when flagging outlier segmentations. It is not effective in determining small changes between images, as it focuses on detecting outlier regions. Future experiments involving artifact corruptions of less severity can provide more information about the method's performance in determining minor segmentation failures. This remains a challenge for automated brain segmentation, and a detailed manual evaluation provides the most reliable results. Software associated with this project is available at <https://github.com/katembartz/SAFE>, which includes a script to automatically process SLANT and SynthSeg outputs and generate a report indicating whether to accept or reject each segmentation. The MDDs generated from the 38,251 subjects indicating normative measures for region volumes and brain occupancy values is also available for both SLANT and SynthSeg.

SLANT was shown to be more robust to image artifacts, where as SynthSeg often failed to generate a brain segmentation when the input image contained simulated artifacts. On the other hand, SLANT segmentation contains inherent randomness that can influence reproducibility, while SynthSeg is deterministic under the same computational environment. The normative measures produced by the two algorithms were only loosely correlated. Therefore, the choice to use SLANT versus SynthSeg as a brain segmentation and parcellation tool should be problem-specific, where each of these factors is considered.

ACKNOWLEDGMENTS

This work was partially supported by the National Science Foundation Graduate Research Fellowships under Grant No. DGE-2139757 (S.P. Hays) and Grant No. DGE-1746891 (S.W. Remedios). The project is also partially supported by CDMRP W81XWH2010912 (J.L. Prince), a grant from the NIH through the NIBIB R01-EB036013 (J.L. Prince), and a Johns Hopkins Discovery Grant (A. Carass).

The authors also acknowledge the publicly available databases used in this study per their respective data-use agreements.

A4: The A4 Study was a secondary prevention trial in preclinical Alzheimer's disease, aiming to slow cognitive decline associated with brain amyloid accumulation in clinically normal older individuals. The A4 Study was funded by a public-private-philanthropic partnership, including funding from the National Institutes of Health-National Institute on Aging, Eli Lilly and Company, Alzheimer's Association, Accelerating Medicines Partnership, GHR Foundation, an anonymous foundation, and additional private donors, with in-kind support from Avid Radiopharmaceuticals, Cogstate, Albert Einstein College of Medicine and the Foundation for Neurologic Diseases. The companion observational Longitudinal Evaluation of Amyloid Risk and Neurodegeneration (LEARN) Study was funded by the Alzheimer's Association and GHR Foundation. The A4 and LEARN Studies were led by Dr. Reisa Sperling at Brigham and Women's Hospital, Harvard Medical School, and Dr. Paul Aisen at the Alzheimer's Therapeutic Research Institute (ATRI) at the University of Southern California. The A4 and LEARN Studies were coordinated by ATRI at the University of Southern California, and the data are made available under the auspices of Alzheimer's Clinical Trial Consortium through the Global Research & Imaging Platform (GRIP). The complete A4 Study Team list is available on: <https://www.actcinfo.org/a4-study-team-lists/>. We would like to acknowledge the dedication of the study participants and their study partners who made the A4 and LEARN Studies possible.

ABIDE: Primary support for the work by Adriana Di Martino was provided by the (NIMH K23MH087770) and the Leon Levy Foundation. Primary support for the work by Michael P. Milham and the INDI team was provided by gifts from Joseph P. Healy and the Stavros Niarchos Foundation to the Child Mind Institute, as well as by an NIMH award to MPM NIMH R03MH096321). Funding sources for the ABIDE dataset are listed at https://fcon_1000.projects.nitrc.org/indi/abide/abide_I.html.

ABIDEII: Primary support for the work by Adriana Di Martino and her team was provided by the National Institute of Mental Health (NIMH 5R21MH107045). Primary support for the work by Michael P. Milham and his team provided by the National Institute of Mental Health (NIMH 5R21MH107045); Nathan S. Kline Institute of Psychiatric Research). Additional Support was provided by gifts from Joseph P. Healey, Phyllis

Green and Randolph Cowen to the Child Mind Institute. Funding sources for the ABIDE II dataset are listed at https://fcon_1000.projects.nitrc.org/indi/abide/abide_II.html.

¹**ADNI:** Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The original goal of ADNI was to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer's disease (AD). The current goals include validating biomarkers for clinical trials, improving the generalizability of ADNI data by increasing diversity in the participant cohort, and to provide data concerning the diagnosis and progression of Alzheimer's disease to the scientific community. For up-to-date information, see adni.loni.usc.edu. Data collection and sharing for the Alzheimer's Disease Neuroimaging Initiative (ADNI) is funded by the National Institute on Aging (National Institutes of Health Grant U19AG024904). The grantee organization is the Northern California Institute for Research and Education. In the past, ADNI has also received funding from the National Institute of Biomedical Imaging and Bioengineering, the Canadian Institutes of Health Research, and private sector contributions through the Foundation for the National Institutes of Health (FNIH) including generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC.; Johnson & Johnson Pharmaceutical Research & Development LLC.; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC.; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics.

Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at: http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf.

²**AIBL:** Data was collected by the AIBL study group. AIBL study methodology has been reported previously (Ellis et al. 2009). Data used in the preparation of this article was obtained from the Australian Imaging Biomarkers and Lifestyle flagship study of ageing (AIBL) funded by the Commonwealth Scientific and Industrial Research Organisation (CSIRO) which was made available at the ADNI database (www.loni.usc.edu/ADNI). The AIBL researchers contributed data but did not participate in analysis or writing of this report. AIBL researchers are listed at www.aibl.csiro.au.

GSP: Data were provided [in part] by the Brain Genomics Superstruct Project of Harvard University and the Massachusetts General Hospital, (Principal Investigators: Randy Buckner, Joshua Roffman, and Jordan Smoller), with support from the Center for Brain Science Neuroinformatics Research Group, the Athinoula A. Martinos Center for Biomedical Imaging, and the Center for Human Genetic Research. 20 individual investigators at Harvard and MGH generously contributed data to the overall project.

³**HABS:** Research reported on this publication was supported by the National Institute on Aging of the National Institutes of Health under Award Numbers R01AG054073, R01AG058533, P41EB015922 and U19AG078109. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. HABS-HD MPIs: Sid E O'Bryant, Kristine Yaffe, Arthur Toga, Robert Rissman, & Leigh Johnson; and the HABS-HD Investigators: Meredith Braskie, Kevin King, James R Hall, Melissa Petersen, Raymond Palmer, Robert Barber, Yonggang Shi, Fan Zhang, Rajesh Nandy, Roderick McColl, David Mason, Bradley Christian, Nicole Philips, Stephanie Large, Joe Lee, Badri Vardarajan, Monica Rivera Mindt, Amrita Cheema, Lisa Barnes, Mark Mapstone, Annie Cohen, Amy Kind, Ozioma Okonkwo, Raul Vintimilla, Zhengyang Zhou, Michael Donohue, Rema Raman, Matthew Borzage, Michelle Mielke, Beau Ances, Ganesh Babulal, Jorge Llibre-Guerra, Carl Hill and Rocky Vi.

HCP1200: Data collection and sharing for this project was provided by the Human Connectome Project (HCP; Principal Investigators: Bruce Rosen, M.D., Ph.D., Arthur W. Toga, Ph.D., Van J. Weeden, MD). HCP

funding was provided by the National Institute of Dental and Craniofacial Research (NIDCR), the National Institute of Mental Health (NIMH), and the National Institute of Neurological Disorders and Stroke (NINDS). HCP data are disseminated by the Laboratory of Neuro Imaging at the University of Southern California. Data used in the preparation of this work were obtained from the Human Connectome Project (HCP) database (<https://ida.loni.usc.edu/login.jsp>). The HCP project (Principal Investigators: Bruce Rosen, M.D., Ph.D., Martinos Center at Massachusetts General Hospital; Arthur W. Toga, Ph.D., University of Southern California, Van J. Weeden, MD, Martinos Center at Massachusetts General Hospital) is supported by the National Institute of Dental and Craniofacial Research (NIDCR), the National Institute of Mental Health (NIMH) and the National Institute of Neurological Disorders and Stroke (NINDS). HCP is the result of efforts of co-investigators from the University of Southern California, Martinos Center for Biomedical Imaging at Massachusetts General Hospital (MGH), Washington University, and the University of Minnesota.

ICBM: Data collection and sharing for this project was provided by the International Consortium for Brain Mapping (ICBM; Principal Investigator: John Mazziotta, MD, PhD). ICBM funding was provided by the National Institute of Biomedical Imaging and BioEngineering. ICBM data are disseminated by the Laboratory of Neuro Imaging at the University of Southern California. Data used in the preparation of this work were obtained from the International Consortium for Brain Mapping (ICBM) database (www.loni.usc.edu/ICBM). The ICBM project (Principal Investigator John Mazziotta, M.D., University of California, Los Angeles) is supported by the National Institute of Biomedical Imaging and BioEngineering. ICBM is the result of efforts of co-investigators from UCLA, Montreal Neurologic Institute, University of Texas at San Antonio, and the Institute of Medicine, Juelich/Heinrich Heine University - Germany.

IXI: IXI data downloaded from <https://brain-development.org/ixi-dataset>.

⁴MCSA: Data used in preparation of this article were shared by the Mayo Clinic Study of Aging (MCSA). The MCSA is funded by the following sources: NIH U01 AG006786, R01 AG034676, R37 AG011378, R01 AG041851, R01 NS097495, R01 AG056366, R01 AG068206, P30 AG062677, GHR Foundation, Elsie and Marvin Dekelboum Family Foundation, Liston Award, Schuler Foundation, Alexander Foundation, Mayo Foundation for Medical Education and Research. The investigators and staff of the MCSA contributed to producing the shared data, but they did not participate in this study's design, its analyses, or the writing of this publication. The MCSA ensures only the scientific integrity of the data that it shared, but they did not review this publication and not responsible for its content.

NACC: The NACC database is funded by NIA/NIH Grant U24 AG072122. NACC data are contributed by the NIA-funded ADRCs: P30 AG062429 (PI James Brewer, MD, PhD), P30 AG066468 (PI Oscar Lopez, MD), P30 AG062421 (PI Bradley Hyman, MD, PhD), P30 AG066509 (PI Thomas Grabowski, MD), P30 AG066514 (PI Mary Sano, PhD), P30 AG066530 (PI Helena Chui, MD), P30 AG066507 (PI Marilyn Albert, PhD), P30 AG066444 (PI David Holtzman, MD), P30 AG066518 (PI Lisa Silbert, MD, MCR), P30 AG066512 (PI Thomas Wisniewski, MD), P30 AG066462 (PI Scott Small, MD), P30 AG072979 (PI David Wolk, MD), P30 AG072972 (PI Charles DeCarli, MD), P30 AG072976 (PI Andrew Saykin, PsyD), P30 AG072975 (PI Julie A. Schneider, MD, MS), P30 AG072978 (PI Ann McKee, MD), P30 AG072977 (PI Robert Vassar, PhD), P30 AG066519 (PI Frank LaFerla, PhD), P30 AG062677 (PI Ronald Petersen, MD, PhD), P30 AG079280 (PI Jessica Langbaum, PhD), P30 AG062422 (PI Gil Rabinovici, MD), P30 AG066511 (PI Allan Levey, MD, PhD), P30 AG072946 (PI Linda Van Eldik, PhD), P30 AG062715 (PI Sanjay Asthana, MD, FRCP), P30 AG072973 (PI Russell Swerdlow, MD), P30 AG066506 (PI Glenn Smith, PhD, ABPP), P30 AG066508 (PI Stephen Strittmatter, MD, PhD), P30 AG066515 (PI Victor Henderson, MD, MS), P30 AG072947 (PI Suzanne Craft, PhD), P30 AG072931 (PI Henry Paulson, MD, PhD), P30 AG066546 (PI Sudha Seshadri, MD), P30 AG086401 (PI Erik Roberson, MD, PhD), P30 AG086404 (PI Gary Rosenberg, MD), P20 AG068082 (PI Angela Jefferson, PhD), P30 AG072958 (PI Heather Whitson, MD), P30 AG072959 (PI James Leverenz, MD).

OASIS-3: OASIS-3: Longitudinal Multimodal Neuroimaging: Principal Investigators: T. Benzinger, D. Marcus, J. Morris; NIH P50 AG00561, P30 NS09857781, P01 AG026276, P01 AG003991, R01 AG043434, UL1 TR000448, R01 EB009352. AV-45 doses were provided by Avid Radiopharmaceuticals, a wholly owned subsidiary of Eli Lilly.

OASIS-4: OASIS-4: Clinical Cohort: Principal Investigators: T. Benzinger, L. Koenig, P. LaMontagne.

PPMI: Data used in the preparation of this article was obtained on 2025-02-20 from the Parkinson's Progression Markers Initiative (PPMI) database (www.ppmi-info.org/access-data-specimens/download-data), RRID:SCR_006431. For up-to-date information on the study, visit www.ppmi-info.org. PPMI – a public-private partnership – is funded by the Michael J. Fox Foundation for Parkinson's Research and funding partners, including 4D Pharma, Abbvie, AcureX, Allergan, Amathus Therapeutics, Aligning Science Across Parkinson's, AskBio, Avid Radiopharmaceuticals, BIAL, BioArctic, Biogen, Biohaven, BioLegend, BlueRock Therapeutics, Bristol-Myers Squibb, Calico Labs, Capsida Biotherapeutics, Celgene, Cerevel Therapeutics, Coave Therapeutics, DaCapo Brainscience, Denali, Edmond J. Safra Foundation, Eli Lilly, Gain Therapeutics, GE HealthCare, Genentech, GSK, Golub Capital, Handl Therapeutics, Insitro, Jazz Pharmaceuticals, Johnson & Johnson Innovative Medicine, Lundbeck, Merck, Meso Scale Discovery, Mission Therapeutics, Neurocrine Biosciences, Neuron23, Neuropore, Pfizer, Piramal, Prevail Therapeutics, Roche, Sanofi, Servier, Sun Pharma Advanced Research Company, Takeda, Teva, UCB, Vanqua Bio, Verily, Voyager Therapeutics, the Weston Family Foundation and Yumanity Therapeutics.

SCAN: SCAN is a multi-institutional project that was funded as a U24 grant (AG067418) by the National Institute on Aging in May 2020. Data collected by SCAN and shared by NACC are contributed by the NIA-funded ADRCs as follows: Arizona Alzheimer's Center - P30 AG072980 (PI: Eric Reiman, MD); R01 AG069453 (PI: Eric Reiman (contact), MD); P30 AG019610 (PI: Eric Reiman, MD); and the State of Arizona which provided additional funding supporting our center; Boston University - P30 AG013846 (PI Neil Kowall MD); Cleveland ADRC - P30 AG062428 (James Leverenz, MD); Cleveland Clinic, Las Vegas – P20AG068053; Columbia - P50 AG008702 (PI Scott Small MD); Duke/UNC ADRC - P30 AG072958; Emory University - P30AG066511 (PI Levey Allan, MD, PhD); Indiana University - R01 AG19771 (PI Andrew Saykin, PsyD); P30 AG10133 (PI Andrew Saykin, PsyD); P30 AG072976 (PI Andrew Saykin, PsyD); R01 AG061788 (PI Shannon Risacher, PhD); R01 AG053993 (PI Yu-Chien Wu, MD, PhD); U01 AG057195 (PI Liana Apostolova, MD); U19 AG063911 (PI Bradley Boeve, MD); and the Indiana University Department of Radiology and Imaging Sciences; Johns Hopkins - P30 AG066507 (PI Marilyn Albert, PhD.); Mayo Clinic - P50 AG016574 (PI Ronald Petersen MD PhD); Mount Sinai - P30 AG066514 (PI Mary Sano, PhD); R01 AG054110 (PI Trey Hedden, PhD); R01 AG053509 (PI Trey Hedden, PhD); New York University - P30AG066512-01S2 (PI Thomas Wisniewski, MD); R01AG056031 (PI Ricardo Osorio, MD); R01AG056531 (PIs Ricardo Osorio, MD; Girardin Jean-Louis, PhD); Northwestern University - P30 AG013854 (PI Robert Vassar PhD); R01 AG045571 (PI Emily Rogalski, PhD); R56 AG045571, (PI Emily Rogalski, PhD); R01 AG067781, (PI Emily Rogalski, PhD); U19 AG073153, (PI Emily Rogalski, PhD); R01 DC008552, (M.-Marsel Mesulam, MD); R01 AG077444, (PIs M.-Marsel Mesulam, MD, Emily Rogalski, PhD); R01 NS075075 (PI Emily Rogalski, PhD); R01 AG056258 (PI Emily Rogalski, PhD); Oregon Health Science University - P30 AG066518 (PI Lisa Silbert, MD, MCR); Rush University - P30 AG010161 (PI David Bennett MD); Stanford – P30AG066515; P50 AG047366 (PI Victor Henderson MD MS); University of Alabama, Birmingham – P20; University of California, Davis - P30 AG10129 (PI Charles DeCarli, MD); P30 AG072972 (PI Charles DeCarli, MD); University of California, Irvine - P50 AG016573 (PI Frank LaFerla PhD); University of California, San Diego - P30AG062429 (PI James Brewer, MD, PhD); University of California, San Francisco - P30 AG062422 (Rabinovici, Gil D., MD); University of Kansas - P30 AG035982 (Russell Swerdlow, MD); University of Kentucky - P30 AG028283-15S1 (PIs Linda Van Eldik, PhD and Brian Gold, PhD); University of Michigan ADRC - P30AG053760 (PI Henry Paulson, MD, PhD) P30AG072931 (PI Henry Paulson, MD, PhD) Cure Alzheimer's Fund 200775 - (PI Henry Paulson, MD, PhD) U19 NS120384 (PI Charles DeCarli, MD, University of Michigan Site PI Henry Paulson, MD, PhD) R01 AG068338 (MPI Bruno Giordani, PhD, Carol Persad, PhD, Yi Murphey, PhD) S10OD026738-01 (PI Douglas Noll, PhD) R01 AG058724 (PI Benjamin Hampstead, PhD) R35 AG072262 (PI Benjamin Hampstead, PhD) W81XWH2110743 (PI Benjamin Hampstead, PhD) R01 AG073235 (PI Nancy Chiaravalloti, University of Michigan Site PI Benjamin Hampstead, PhD) 1I01RX001534 (PI Benjamin Hampstead, PhD) IRX001381 (PI Benjamin Hampstead, PhD); University of New Mexico - P20 AG068077 (Gary Rosenberg, MD); University of Pennsylvania - State of PA project 2019NF4100087335 (PI David Wolk, MD); Rooney Family Research Fund (PI David Wolk, MD); R01 AG055005 (PI David Wolk, MD); University of Pittsburgh - P50 AG005133 (PI Oscar Lopez MD); University of Southern California - P50 AG005142 (PI Helena Chui MD); University of Washington - P50 AG005136 (PI Thomas Grabowski MD); University of Wisconsin - P50 AG033514 (PI Sanjay Asthana MD FRCP); Vanderbilt University – P20 AG068082; Wake Forest - P30AG072947 (PI Suzanne Craft, PhD); Washington University, St. Louis - P01 AG03991 (PI John Morris MD); P01 AG026276 (PI John Morris MD); P20 MH071616 (PI Dan Marcus); P30 AG066444 (PI John Morris MD); P30 NS098577

(PI Dan Marcus); R01 AG021910 (PI Randy Buckner); R01 AG043434 (PI Catherine Roe); R01 EB009352 (PI Dan Marcus); UL1 TR000448 (PI Brad Evanoff); U24 RR021382 (PI Bruce Rosen); Avid Radiopharmaceuticals / Eli Lilly; Yale - P50 AG047270 (PI Stephen Strittmatter MD PhD); R01AG052560 (MPI: Christopher van Dyck, MD; Richard Carson, PhD); R01AG062276 (PI: Christopher van Dyck, MD); Florida - P30AG066506-03 (PI Glenn Smith, PhD); P50 AG047266 (PI Todd Golde MD PhD).

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